SC42101 Networked and Distributed Control Systems

Assignment 1 - Lectures 1 and 2

Instructions:

- The assignments are individual. You may consult and discuss with your colleagues, but you need to provide independent answers.
- You may use Matlab/Python/Julia to solve the assignment. Many questions require you clearly to do so.
- Provide detailed answers, describing the steps you followed.
- Provide clear reports, typed, preferably using LaTex.
- Include with your report the **"in-depth" AI acknowledgement** from the ME Faculty. Not doing so will result in your report not being considered handed on time.
- Submit the reports digitally on Brightspace as indicated on the lectures.
- Include your code, for reproducibility of your results, in your Brightspace submission in a zip file, or through a link on your report. Make sure the report is in a separate PDF.
- The code will *only* be used to verify reproducibility in case of doubts. The grading will be performed based on the results described in the report.

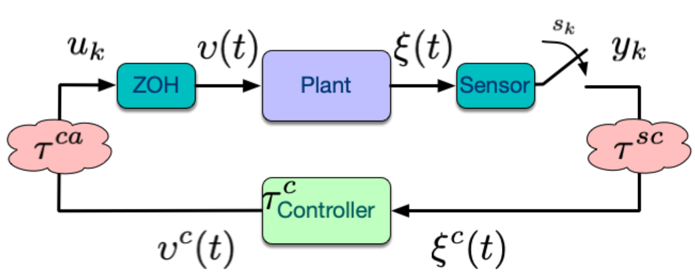

Figure 1: Schematic of a NCS including delays.

Consider a sample-data networked control system (NCS), as depicted in Figure 1. Assume that the plant dynamics are linear time-invariant given by:

$$\dot{\xi}(t) = A\xi(t) + Bv(t),$$

where the control signal v is a piece-wise constant signal resulting from the application of a controller in sampled-and-hold fashion, i.e. $v(t) = u_k,\ t \in [s_k,\ s_{k+1})$.

The system matrices are given by:

$$A = \begin{bmatrix} 0 & 0.5 + c \\ 0.5 + |a - b| & 1 \end{bmatrix},\ B = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

where a is given by the first digit, b by the third digit, and c by the last digit of your student ID number.

We start considering the system with a constant sampling interval $h = s_{k+1} - s_k$ for all k, and assuming no delay is present $\tau = \tau^{sc} + \tau^{c} + \tau^{ca} = 0$.

Important note: For all the questions that follow, you do **NOT** necessarily need to provide analytic expressions, unless explicitly stated. However, you need to describe the steps you took to produce the resulting plots, and design process you followed.

Question 1: (1p)

1. (1p) Design a linear continuous time controller $v(t) = -K\xi(t)$ placing the poles of the continuous closed-loop at $-2 + j$ and $-2 - j$. Construct the exact discrete-time model $x_{k+1} = F(h)x_k + G(h)u_k$, and the closed-loop system resulting from applying the controller in sample-data (piece-wise constant) fashion: $u_k = -Kx_k$, $x_k := \xi(s_k)$. Select sufficiently many sampling times h to provide ranges of sampling times for which the sampled-data system is stable. Comment on what you observe.

Next we consider the case when delays are present in the networked system. Assume that the system is affected by a **constant** delay $\tau \in [0, 2h)$, and it is controlled with the *static* controller K.

Assume a mechanism in which the controller holds the last input computed **but** reset the input to zero if at an update time s_k the applied input is older than 2 samples. That is, in the time interval $[s_k, s_{k+1}]$ the controller is **never** applying an input u_r where $r \leq k - 2$.

Question 2: (4p)

1. (2p) Derive the exact discrete-time model for the NCS with delays under the descrbed mechanism.

2. (1p) Study by testing on sufficiently many combinations (h, τ) of sampling intervals and system delays that result in an asymptotically stable closed-loop. Provide plots illustrating the combinations of (h, τ) retaining stability.

3. (1p) Select a sampling interval h that guarantees stability under no delays. Redesign the controller to improve the robustness against delays, that is, try to design a controller that increases the range of tolerable delays for the selected h. Think about designing now a *dynamic* controller instead.

Now we assume that the delays can take two different values $\tau_1 < h$ and $\tau_2 \in [h, 2h)$.
Question 3: (5p)

1. (1p) Without assuming knowledge of the sequence of delays, describe the set of LMIs that need to be solved to determine if the static controller K designed in Question 1 guarantees stability for this NCS.
2. (2p) Assume now that the possible sequences of delays is restricted to be either[1]: $(\tau_1\tau_1\tau_2)^\omega$ or $(\tau_2)^\omega$. Can you simplify the check required to test stability of a given sampled-data controller in this case? e.g. can you reduce the size of the LMIs, if at all needed?
3. (2p) Under the same scenario as in the previous question. Find values of h, τ_1 and τ_2 guaranteeing the stability of the closed loop system. Verify and illustrate the result through some simulations (show at least a stable and an unstable simulation trajectory).

[1]The notation $(a)^\omega$ denotes the infinite repetition of that symbol, in the two cases in the question that means that the sequences end-up repeating forever the last inter-sample time